

Class 24 Polynomials

Goals

1. Know how real numbers are categorized as algebraic or transcendental.
2. Be able to prove that a given number is algebraic over the integers or the rationals.
3. Know the statement of the fundamental Theorem of Algebra.
4. Find zeros of given polynomials.

We are familiar with the fact that the real numbers can be partitioned into disjoint subsets of rationals and irrationals.

Another way to categorize real numbers is in terms of zeros of polynomials with rational coefficients.

Terminology:

Notation:

Examples:

Poly of degree 5 over $\mathbb{R}$.

Quadratic poly in $\mathbb{C}[x]$ but not in $\mathbb{R}[x]$

Cubic poly in $\mathbb{Z}[x]$

Degree 10 poly in $\mathbb{Z}_6[x]$

Definition: A zero for $p(x)$ over F is

Examples: Verify that $2 + i$ and $2 - i$ are zeros $p(x) = x^2 - 4x + 5$.

Complex Conjugate Theorem:

Examples:

1. find all zeros for $x^2 + 1$ over $\mathbb{R}$.

2. Find all zeros for $x^2 + 1$ over $\mathbb{C}$.

3. Find all zeros for $x^2 + 1$ over $\mathbb{Z}_5$.

Definition: An algebraic number is

Examples:

Prove $\sqrt{2}$ is algebraic.

Prove $\sqrt{2} + \sqrt{3}$ is algebraic.

